

Varun Prasad Muchanapally

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SUMMARY

AI Engineer specializing in building scalable RAG systems, LLM-powered applications, and end-to-end machine learning pipelines using Python, LangChain, and cloud platforms.

EDUCATION

University of Central Florida | *M.S., Computer Science* (GPA: 3.9/4)

Orlando, FL Aug 2024 - May 2026

B V Raju Institute of Technology | *B.Tech., Computer Science* (GPA: 8.05/10)

Hyderabad, India Aug 2020 - Jun 2024

PROJECTS

Hermes - Hurricane Simulator

- Built a **hurricane evacuation simulator** using **Google ADK**, **Gemini AI**, and **Mapbox** where AI agents help people escape storms safely.
- Used **Agent2Agent (A2A)** so agents talk to each other and pick the safest evacuation zones based on the storm's real path and **Saffir-Simpson** wind physics.
- Used **FastAPI** and **asyncio** to run 80 civilian AI agents in parallel, each making their own evacuation decision shown live on an interactive map.

GitHub Diagnostics Checker

- Built an AI Agent using **LangGraph** that scans GitHub repos, finds bugs, and autonomously opens PRs with fixes via **PyGithub**
- React** and **WebSockets** power a live dashboard that streams real-time agent status updates as the agent scans and fixes repos
- Used **FastAPI** as the backend with **APScheduler** to trigger scheduled autonomous diagnostics, eliminating the need for manual code reviews

AI-RAG Quiz Application

- Built a quiz website using **Python**, **FastAPI**, and **React** where users upload a PDF and get an AI-generated quiz
- Used **RAG** with **Pinecone** and **GPT-4o-mini** with **Multi-Query Retrieval** and **semantic chunking** to retrieve relevant chunks.
- Deployed on **GCP Cloud Run** with a **GitHub Actions** CI/CD pipeline, with **MD5-based caching** to eliminate redundant LLM calls and reduce API costs.

Personal AI OS - Multimodal Personal Knowledge Assistant

- Built a personal AI assistant using **Google ADK** that connects to **Gmail**, **Google Calendar** and **Google Docs** etc.
- Used **Pinecone** as the vector store with **Multi-Query Retrieval**, **semantic chunking** and **reranking** to search across all personal data and return the most accurate answer
- Built an **agentic pipeline** that automatically syncs new emails and calendar events in real time keeping the knowledge base always up to date

EXPERIENCE

One Convergence | *Software Engineer*

Apr 2023 - Nov 2023

- Integrated OpenAI embeddings (text-embedding-ada-002) to convert 10,000+ product descriptions into vectors, enabling AI-powered semantic search that improved result relevance
- Weaviate Vector DB** stored and queried embeddings as a semantic search engine, delivering higher relevance than keyword-based search
- Selenium** automated URL validation across 10,000+ product listings, eliminating broken links going unnoticed

Forage AI | *Python Developer (Web Scraping/ Crawling)*

Mar 2022 - Dec 2022

- Automated legitimacy verification using Selenium across 100,000+ URLs, eliminating manual site reviews
- Scraped and extracted structured data from 200+ unstructured websites using BeautifulSoup, feeding the data pipeline and enabling timely analytics
- Python** scripts logged 200+ weekly data quality issues surfaced from scraped datasets reducing review time for the team

TECH STACK

- Languages:** Python, Java, TypeScript, SQL, C, C++, JavaScript, R, HTML/CSS
- AI & LLM Systems:** LLMs, RAG, AI Agents, LangChain, LangGraph, LlamaIndex, Hugging Face, Transformers, PyTorch, Prompt Engineering, LoRA, QLoRA (PEFT), LangSmith, Accelerate, Hugging Face Datasets
- Vector Databases & Retrieval:** FAISS, Pinecone, Weaviate, ChromaDB
- Backend & APIs:** FastAPI, Flask, Node.js, Express.js, REST APIs, Microservices, Postman
- Frontend & App Development:** React, Tailwind CSS, Bootstrap
- Cloud & AI Platforms:** AWS
- Data & Databases:** PostgreSQL, MySQL, MongoDB, Redis, ETL Pipelines, Power BI
- DevOps & Systems:** Docker, Kubernetes, CI/CD (GitHub Actions), Distributed Systems